

Simulate Your PIC Microcontroller Using **gpsim**

In this article, we explore the features and advantages of **gpsim** by analysing the step-by-step procedure to simulate PIC microcontroller using this tool. **gpsim** is available for Linux operating system on the DVD accompanying this month's issue of **EFY Plus**

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gpsim is a free software simulation tool for PIC microcontrollers from Microchip, which has been designed to be fast enough to simulate PICs at a speed of about 20MHz. Its accurate design makes it comparable to using a real PIC microcontroller from the core to the I/O pins and even all the internal peripherals.

Why it is noteworthy

It is mostly written in C++, which enables it to easily implement a hierarchical model of a PIC. This in turn enables it to use the inheritance property of C++, which allows the various models

Three important functions

1. Complex debugging with in-circuit emulators
2. Provides a means to access virtual functions like execute and name
3. Sets read and write break points on values

Competitors to gpsim

- Real PIC Simulator
- PICsim
- Oshon Simulator
- PIC Development Studio

System requirements

1. No minimum hardware requirement
2. Two packages, readline and gtk (the gimp tool kit), are pre-installed in a Linux distribution like Ubuntu
3. Utility package, gputils

of PIC to share a relationship. By being designed to support all the three families of PIC (12-bit core, 14-bit core and 16-bit core), this tool eliminates the need of a different simulator for a different core. **gpsim**'s accuracy allows it to test the PIC in the same way as it would be in a real world.

Why use this tool

Currently, there are so many software simulators available for PIC that it

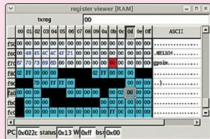


Fig. 1: Register viewer

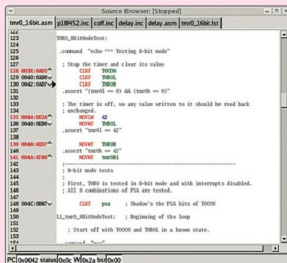


Fig. 2: Source browser

becomes obvious for engineers to ask: Why use **gpsim**? A simple answer to this is that, being a free and open source software, unlike most of its serious competition, **gpsim** enjoys the FOSS community support.

gpsim has the following benefits that make it superior to all the other simulators available in the market:

Easy debugging with three-control option. **gpsim** is capable of being controlled from either a graphical user interface (GUI) by a remote process or by a command line interface (CLI). Although the tool is by default configured for GUI, it can also be configured for CLI and remote process later.

No hardware requirements. **gpsim** does not require any minimum hardware specification and so can be built and run on any popular Linux distribution. Moreover, it is not limited to

Linux and has also been ported to other operating systems like Microsoft Windows, BSD, MAC and Solaris.

Supports loadable modules. This feature enables **gpsim** to add all other devices that are totally separate from this tool. It allows users to customise a simulation environment.

Allows easy inspection. At times, a history or a trace of the work helps inspect its current status and to analyse the